

Tirth Laheri

+1 (615)-497-1076 | tirthlaheri@gmail.com | linkedin.com/in/tirthlaheri | github.com/Tirth1411

EDUCATION

Middle Tennessee State University

Murfreesboro, TN

Master of Science in Data Science; GPA: 4.00/4.00

2024 – Present

- Relevant Coursework: Machine Learning, Statistical Methods, Data Mining, Data Visualization

Parul University

India

Bachelor of Technology in Computer Science (IoT); GPA: 9.06/10.00

2020 – 2024

- Relevant Coursework: Data Structures, Algorithms, Database Systems, Cloud Computing

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, R, JavaScript, HTML/CSS

AI/ML Concepts: Machine Learning, Deep Learning, Generative AI, LLMs, RAG, Prompt Engineering, Computer Vision, NLP, Neural Networks, Transfer Learning, Model Optimization, Statistics, Data Analysis

Frameworks & Tools: TensorFlow, PyTorch, Keras, Scikit-Learn, HuggingFace, LangChain, LlamaIndex, OpenCV, FastAPI

Developer Tools: Git, Docker, Kubernetes, VS Code, Jupyter Notebook, Google Colaboratory, MLflow, Power BI, Tableau, Streamlit

Libraries: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, FAISS

Cloud & Databases: AWS, Cloud Computing, MySQL, PostgreSQL, Vector Databases, ETL Pipelines

EXPERIENCE

Graduate Assistant (Administrative)

Aug 2025 – Present

Physician Assistant Department, Middle Tennessee State University

Murfreesboro, TN

- Built **data pipelines and dashboards** using Python, SQL, Power BI, and Tableau to automate reporting and compliance tracking.
- Prototyped an **LLM-based assistant** (LangChain) to answer staff queries from internal documentation, reducing manual lookups.
- Performed **data analysis and validation** with Pandas/NumPy, improving reporting accuracy and reducing manual effort.
- Collaborated with cross-functional teams using Git for version control and process automation.

AI/ML Intern

Mar 2024 – July 2024

NeuralForge AI

India

- Built and deployed **ML and GenAI models** (TensorFlow, PyTorch, HuggingFace) for predictive analytics and automation.
- Fine-tuned **LLMs** and built **RAG pipelines** (LangChain, vector databases) for internal Q&A and document search use cases.
- Developed **end-to-end ML pipelines**: preprocessing, EDA, feature engineering, training, and evaluation.
- Owned **model deployment and monitoring** using AWS, Docker, MLflow, and REST APIs as part of MLOps workflows.
- Collaborated in **KT sessions** on enterprise GenAI architectures and scalable model management practices.

Data Science Intern

Dec 2023 – Mar 2024

SlashMark

India

- Built **ML pipelines** (Pandas, NumPy, Scikit-learn) for predictive analytics with feature engineering and EDA.
- Trained **supervised models** (Random Forest, Neural Networks) achieving up to 90% accuracy.
- Created **visual analytics reports** (Matplotlib, Seaborn) to communicate insights for business decisions.

PROJECTS

- Multilingual Legal Document Intelligence System** | *GenAI, RAG, OCR, LLM* Jan 2026 – Present
- Building a **RAG pipeline** (FAISS, LangChain) for multilingual legal document search and Q&A.
 - Using **PaddleOCR** for layout-aware extraction and **FastAPI** for scalable inference endpoints.
- GravWaveFormer** | *Deep Learning Ensemble for Gravitational Wave Detection* Sep 2025 – Dec 2025
- Built a **CNN-Transformer + GraphSAGE ensemble** achieving 0.934 ROC-AUC with under 2ms inference latency.
 - Added **explainability** (GradCAM, ablation studies) and an interactive Streamlit dashboard.
- Cross-Camera Player Mapping** | *YOLOv11, Deep SORT, ResNet* July 2025
- Built a **CV tracking pipeline** with YOLOv11 + Deep SORT and ResNet-based Re-ID, reaching 95% cross-camera ID consistency.
- Plant Disease Detection** | *TensorFlow, OpenCV, CNN* Dec 2024
- Built a **CNN classifier** with transfer learning for plant disease identification from leaf images.
- Diabetic Retinopathy Screening** | *TensorFlow, Deep Learning, CNN* Feb 2024
- Built a **deep learning model** for retinopathy detection achieving 94.7% accuracy on medical images.
- Emotion Detection** | *TensorFlow, Facial Recognition, CNN* Dec 2023
- Built a **CNN-based facial emotion classifier** for real-time emotion recognition.
- Speech-To-Speech Multilingual Translation** | *HuggingFace, Google API* Mar 2023
- Built a **real-time speech translation** app with voice cloning for Indian languages.

CERTIFICATIONS

Python for Machine Learning & Data Science – Udemy Certification
Machine Learning Specialization – Coursera Certification
Machine Learning Fast-Track Workshop – Google Certification
Cloud Computing – Parul University
Introduction to Internet of Things – NPTEL Certification
Web Development – 1Stop.ai

PUBLICATIONS

- Visible Light Communication Based Indoor Positioning System** 2024
IEEE Xplore Digital Library
- Built a **VLC-based indoor localization system** using photodiode sensors and **ML-driven calibration**. [Link](#)
- IoT-Based Remote Health Monitoring for Rural Clinics** 2023
Private Research Repository, Parul University
- Designed an **IoT patient monitoring system** with MQTT data transmission and rule-based anomaly detection.
- Smart Wearables and Cloud Computing for Preventive Healthcare** 2022
Private Academic Repository, Parul University
- Built a **wearable-cloud architecture** with AI-driven alerts for chronic disease risk detection.